

Lab6 report_yta92

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Executive Summary

This report presents the findings of a digital forensic investigation conducted on a disk image obtained from a Dell Latitude CPi laptop (Serial No. VLQLW), acquired on September 22, 2004, by Examiner Shane Robinson.

The analysis was performed on an EnCase 4 (E01) forensic image using tools including ewfinfo, mmls, fsstat, and regripper. The integrity of the evidence was verified through MD5 hash validation (aee4fcd9301c03b3b054623ca261959a).

Investigation findings indicate that the primary user account, "Mr. Evil," engaged in a series of deliberate and systematic activities consistent with network reconnaissance, credential theft, and evidence concealment between August 19 and 27, 2004. Key findings include the installation and execution of multiple hacking-related tools (Cain & Abel, Ethereal, NetStumbler), access to a remote FTP server (4.12.220.254), and the use of anonymization tools to obscure activity.

Based on the evidence collected and analyzed, it is concluded that the user was engaged in activities aimed at unauthorized network intrusion and data theft.

1. The Evidence File and Disk Image**1A what, who, where, when, how the evidence file was created.**

Item	Details
Acquisition Date	September 22, 2004 – 07:06:04
Examiner	Shane Robinson
Case Number	NIST CFReDS – Hacking Case
Target Device	Dell Latitude CPi (sn#: VLQLW, hdsn#: RQQF7429)
File Format	EnCase 4 (E01)
Size	Approx. 4.5 GiB (4,871,301,120 bytes)
OS / Software Used	Windows XP / Version 4.19a
Compression Method	deflate (no compression)
Hash (MD5)	aee4fcd9301c03b3b054623ca261959a

✂ This information is obtained from ewfinfo.

✂ The registered owner of the device is identified as Greg Schardt, based on the Windows installation username and registry records.

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1B & C The filesystem and partition of the file system.

```

└─$ sudo mmls /mnt/ewf/ewf1
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length    Description
000:  Meta      0000000000  0000000000  0000000001  Primary Table (#0)
001:  _____ 0000000000  0000000062  0000000063  Unallocated
002:  000:000  0000000063  0009510479  0009510417  NTFS / exFAT (0x07)
003:  _____ 0009510480  0009514259  0000003780  Unallocated

└─$ sudo fsstat -o 63 /mnt/ewf/ewf1
FILE SYSTEM INFORMATION

File System Type: NTFS

```

The partition in slot 002 starts at sector 63 and ends at sector 9510479. It contains an **NTFS file system**, with a partition length of 9510417 sectors, which corresponds to approximately 4.87 GB of data stored (9510417×512 bytes).

```

└─$ sudo file -s /mnt/ewf/ewf1
[sudo] password for yta92:
/mnt/ewf/ewf1: DOS/MBR boot sector MS-MBR XP english at offset 0x12c "Invalid partition table" at o
ffset 0x144 "Error loading operating system" at offset 0x163 "Missing operating system", disk signa
ture 0xec5dec5d; partition 1 : ID=0x7, active, start-CHS (0x0,1,1), end-CHS (0x24f,254,63), startse
ctor 63, 9510417 sectors

```

The disk seems to have boot issues, likely due to a corrupted partition table or boot sector. Repairing the partition table or the OS might be necessary.

2. The System

2A Find out the operating system version, product, etc.

```

└─$ regripper -r software -p winver
Launching winver v.20200525
winver v.20200525
(Software) Get Windows version & build info

ProductName      Microsoft Windows XP
BuildLab         2600.xpclient.010817-1148
RegisteredOrganization  N/A
RegisteredOwner  Greg Schardt
InstallDate      2004-08-19 22:48:27Z

```

2B Find out about the user accounts and profiles. Find out who have logged in to the system

- 5 user accounts confirmed
- **Only Mr. Evil** appears to be active

Item	Explanation
Mr. Evil Account	Has administrative rights and is the most frequently logged-in user. It's a key target for investigation.

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Item	Explanation
Administrator	Exists but not used . Attackers often use this account, but it seems unused here.
Guest Account	Not used , but the password is not required , which is a security risk. Should be disabled.
SUPPORT, HelpAssistant	These accounts are not used , but service accounts like this should be disabled or removed.
Group Membership	Mr. Evil is part of the Administrators group , meaning they have unrestricted access to the system.

※ This information is obtained from samparse.

3. The User Artifact

3A \$ Uninstall: regripper -r software -p uninstall

Tool Name	Type	Installation Date (UTC)
Cain & Abel v2.5 beta45	Password cracking	2004-08-20 15:05:58Z
Anonymizer Bar 2.0 (remove only)	Anonymizer tools	2004-08-20 15:05:09Z
mIRC	Potentially used for C2 communication	2004-08-20 15:10:04Z
123 Write All Stored Passwords	Password recovery/extraction	2004-08-20 15:13:08Z
Look@LAN 2.50 Build 29	Network scanning	2004-08-25 15:56:11Z
Network Stumbler 0.4.0 (remove only)	Wi-Fi scanning	2004-08-27 15:12:15Z
WinPcap 3.01 alpha	Packet capture driver	2004-08-27 15:15:19Z
Ethereal 0.10.6	Network packet analysis	2004-08-27 15:29:19Z

User Activity History and Purpose Inference

Many tools related to security, network analysis, and anonymization were found. From this, it can be inferred that the user may have been using tools for network analysis and possibly hacking.

3B \$ Recentdocs: regripper -r NTUSER.DAT -p recentdocs

Recently Used Files by Extension

.bmp file: *yng13.bmp* (image file)

.rtf file: *Receipt.rtf* (receipt)

.txt files: *channels.txt* and *keys.txt* (text files)

Folder Access

- *Temp* on *m1200* (4.12.220.254) (network drive)
- *channels*, *GhostWare*, *Anonymizer* (local folders)

Network Access (NetHood)

- \\4.12.220.254\Temp (access to a shared folder on IP address 4.12.220.254)

Inferred User Activities

Use of Network Resources:

- The user frequently accessed the *Temp* folder on a machine named *m1200* (4.12.220.254).

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Security-Related Activities:

- Files like *keys.txt* and a folder named *Anonymizer* (possibly an anonymization tool) suggest involvement in some kind of security-related work.

Multimedia File Editing:

- There is evidence of handling image and document files such as *.bmp* and *.rtf* files.

Frequent Use of Text Files:

- The user often edited or referred to files like *channels.txt* and *keys.txt*.

Summary

The user appears to have been engaged in accessing network drives, manipulating security-related files (such as *keys.txt*), and editing image and document files. The presence of files like *keys.txt* and a possibly anonymization-related folder (*Anonymizer*) suggests involvement in security evasion or unauthorized network activities, warranting further investigation.

3C \$ Comdlg32: regripper -r NTUSER.DAT -p comdlg32**User:** Mr. Evil**Period:** August 25–27, 2004

Activity: Frequent execution of network analysis tools (Ethereal, WinPcap, NetStumbler) from the desktop.

Evidence: Registry entries from ComDlg32\LastVisitedMRU and OpenSaveMRU indicate repeated access to .exe files for network tools from the desktop.

Key Files:

- ethereal-setup-0.10.6.exe
- WinPcap_3_01_a.exe
- netstumblerinstaller_0_4_0.exe
- lalsetup250.exe

Implications:

- Indicates deliberate preparation for network reconnaissance.
- Suggests the user organized and launched tools directly from the desktop, reflecting focused and purposeful activity.
- Toolset matches known capabilities for packet sniffing and wireless network scanning.

3D \$ Typedurls: regripper -r NTUSER.DAT -p typedurls

history of URLs manually entered in Internet Explorer (IE).

Order	URL	Notes & Analysis
1	http://www.maktoob.com/	Arab portal site
2	http://www.ethereal.com/	Network packet analysis tool (now Wireshark)
3	http://www.wardriving.com/	Wardriving (wireless LAN exploration) related site
4	http://4.12.220.254/temp/	Local or external IP web server
5	ftp://4.12.220.254/temp	FTP connection to the same server
6	ftp://4.12.220.254	Root FTP directory of the same host
7	http://www.drudgereport.com/	Political news site

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Order	URL	Notes & Analysis
8	http://www.majorgeeks.com/	Freeware and utility distribution site
9	http://www.yahoo.com/	General portal site
10	http://www.2600.org/	Hacker culture-related site
11	http://www.microsoft.com/isapi/redir.dll?...	Microsoft redirect link

Security tools and hacker-related URLs are prominent

- URLs like ethereal.com, wardriving.com, and 2600.org stand out, which are related to network analysis, hacking, and security.

Attempt to access FTP

- ftp://4.12.220.254 indicates a potential connection to either a personal server or an internal organizational server, which could be a subject of investigation.

Possibly high technical literacy

- The use of sites like MajorGeeks and Ethereal suggests the individual may have higher technical knowledge than an average user.

Possible interest in the Middle East region

- The presence of maktoob.com implies potential interest or connections to the Middle East.

3E \$ Usb: regripper -r system -p usb

```
ROOT_HUB [2004-08-19 16:56:14Z]
S/N: 4&15736a3f&0 [2004-08-27 15:08:10Z]
```

- **ROOT_HUB [2004-08-19 16:56:14Z]**
→ The USB hub was first recognized on August 19, 2004, so the USB ports likely became operational starting from that date.
- **S/N: 4&15736a3f&0 [2004-08-27 15:08:10Z]**
→ This indicates that a device with the serial number 4&15736a3f&0 was connected or used at 15:08:10 (UTC) on August 27, 2004.

3F \$ Userassist: regripper -r NTUSER.DAT -p userassist

From the UserAssist key records, it was confirmed that Mr. Evil launched multiple hacking-related tools between August 19 and 27, 2004.

Tool Executions from August 20 to 27:

- **Network Analysis:** Ethereal, WinPcap, Look@LAN
- **Password Cracking:** Cain & Abel
- **Wireless Network Scanning:** NetStumbler
- **Anonymization Tool:** Anonymizer
- **Remote Access:** telnet.exe
- **Possible Botnet Communication:** mIRC
- **Others:** Faber Toys, Agent, Whois

Many of these executables were launched from the Desktop or Temp folder, suggesting that the activities were deliberately prepared. Notably, on **August 27**, several tools were executed in rapid

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succession (Ethereal, Cain, NetStumbler, etc.), which is believed to represent the **final stage of a network intrusion**.

3G \$ Shimcache: regripper -r system -p shimcache**Traces of Suspicious or Tool-related Files**

File Path	Description
C:\Program Files\Cain\Cain.exe	Password sniffer. Used in both ethical hacking and malicious activities.
WinPcap_3_0_1a.exe, npf_mgm.exe, daemon_mgm.exe	Packet capture drivers. May be used in conjunction with Cain.
mIRC\mirc.exe	Chat client, but can be misused for botnet communications.
NetStumbler.exe, netstumblerinstaller_0_4_0.exe	Wi-Fi sniffer. Wireless network reconnaissance tool.
1L93GJAJ.bat	Batch file. Likely an auto-executing script, possibly manually created or malware-generated.
lalsetup250.exe	Some kind of setup file, but details are unclear (suspicious due to randomized filename).
ANONYM~1.DLL	Anonymization tool (possibly related to anonymous browsing?).

Inference of User Behavior

- **Network reconnaissance activities:** Packet analysis (Ethereal), credential harvesting (Cain), wireless network investigation (NetStumbler)
- **Anonymization attempts:** Use of anonymous browsing tools

Possible User Actions in Chronological Order

- Many files were recorded between **August 20 and 27, 2004**, suggesting that the user or malware was actively performing operations during this period.

3H \$ Runmru: regripper -r NTUSER.DAT -p runmru

```
LastWrite Time 2004-08-26 15:05:15Z
MRUList = dcba
a  www.cnn.com\1
b  cmd\1
c  www.google.com\1
d  telnet\1
```

Suspicious Execution Commands:

- Telnet may have been used by an attacker for remote access.
- Direct execution of a website may be associated with traces of malware or shortcut-based attacks (such as .url files).

4. Conclusion

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Based on a comprehensive analysis of the following information, it is highly likely that a user named "Mr. Evil" was engaged in activities aimed at network intrusion and security breaches in August 2004. Below is a summary of specific evidence and the suspected actions:

User Profile and Behavioral Characteristics**Technical Skills**

- Frequently used network analysis tools (Ethereal/Wireshark, WinPcap), password cracking tools (Cain & Abel), and Wi-Fi scanning tools (NetStumbler).
- Command-line operations (cmd), remote access via telnet, and connection history to an FTP server (4.12.220.254).
- Access to anonymization tools (Anonymizer) and hacker-related websites (2600.org).

Timeline of Activities

- Concentrated installation and execution of tools from August 19 to 27, 2004 (determined from timestamps in ShimCache, RunMRU, and UserAssist).
- On August 27 in particular, records show USB device connections and access to a network drive (\\4.12.220.254\\Temp).

Suspicious Files and Tools

- keys.txt (likely containing passwords or encryption keys), 1L93GJAJ.bat (an autorun script), lalsetup250.exe (an unidentified executable).
- Malware-like behavior, such as the presence of mIRC, which could be used for botnet communication.

Suspected Malicious Activities**Network Reconnaissance**

- Packet sniffing using Ethereal/WinPcap and scanning wireless networks using NetStumbler.
- FTP connection to the IP address 4.12.220.254 suggests possible data exfiltration or use as a staging server for attacks.

Credential Theft

- Attempted password hash cracking with Cain & Abel.
- Sensitive information may have been stored in keys.txt or Receipt.rtf.

Evidence Concealment

- Use of anonymization tools (Anonymizer) and temporary folders (Temp) to hide activities. A user named "Mr. Evil" appears to have been in an advanced attack phase, utilizing sophisticated techniques aimed at network intrusion and data theft.